

Proposing a hybrid deep learning model with sentiment and historical data for cryptocurrencies price prediction

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ABSTRACT

The rapid growth and fluctuations in the cryptocurrency market have attracted significant attention in the development of predictive models. However, many existing studies rely solely on historical price data or use single deep learning models, which often fail to capture the complex and dynamic nature of the market. Moreover, the limited use of sentiment information from news sources has restricted prediction accuracy in previous work. To address these limitations, this article investigates a hybrid deep learning approach that combines CNN and GRU to enhance the accuracy and reliability of predicting the prices of Bitcoin (BTC), Ethereum (ETH), and Monero (XMR). The proposed model integrates both historical price data and Google news sentiment, extracted using widely used tools such as Flair, Vader, and TextBlob. Extensive experiments on diverse datasets demonstrate that the model outperforms existing approaches across multiple evaluation metrics (MAE, MAPE, MSE, RMSE). By simultaneously leveraging spatial and temporal features, the framework captures complex patterns and adapts effectively to market fluctuations. The findings confirm that integrating sentiment analysis with deep learning provides a more robust solution for cryptocurrency price prediction compared to prior studies.

1. Introduction

Financial markets refer to systems and platforms where individuals, companies, and governments can buy and sell financial assets such as stocks, bonds, commodities, currencies, and derivatives (Hull, 2009). One of these financial markets is cryptocurrency market, which is currently one of the hottest topics. Cryptocurrencies are digital forms of money that are stored and exchanged through an independent computer network, separate from banks, governments, or any central authority (Rathee et al., 2023). These currencies are organized by blockchain and are used as a ledger for transactions, including transfers, purchases, or sales. The decentralized nature, immutability, and transparency of blockchain and cryptocurrencies have contributed to their increasing popularity. One of the reasons that has made cryptocurrencies a suitable investment option is the upward trend and the proliferation of exchanges. The most famous cryptocurrency, Bitcoin (BTC), was originally introduced in 2009 and was the only blockchain-based cryptocurrency for over two years (Bitcoin, 2008). However, today there are more than twenty thousand encrypted currencies. Bitcoin has recently gained

significant attention in the fields of economics, cryptography, and computer science (Hamayel & Owda, 2021). The Bitcoin network maintains a record of transactions between individuals, with each record being encrypted, and each record including the encrypted hash of the previous block (Jiang, 2020). The cryptocurrency market is characterized by high volatility and frequent fluctuations. To achieve substantial profits within this environment, continuous monitoring and access to extensive datasets are essential. Moreover, due to the market's inherently complex patterns, the development of robust predictive models capable of accurately forecasting price movements holds significant importance.

Artificial intelligence refers to a set of techniques and algorithms used to program computers in a way that they can perform tasks without the need for human intervention. The main objective of artificial intelligence is to create and enhance intelligent capabilities of machines. One of the important and popular concepts in artificial intelligence is deep learning algorithms (Fetzer, 1990). One of the key characteristics of deep learning algorithms is their ability to recognize complex patterns while simultaneously accounting for multiple factors. Before the use of

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deep learning algorithms for predicting the prices of cryptocurrencies, traditional machine learning algorithms such as ARIMA, regression algorithms, decision trees, random forests, and gradient boosting were used (Derbentsev et al., 2020; Oyedele et al., 2023; Pekel, 2020; Peng et al., 2018; Phaladisailoed & Numnonda, 2018). Cryptocurrencies demonstrate highly complex patterns, making price prediction a challenging time series problem. Numerous research studies have employed deep learning algorithms for this task, owing to their enhanced capability to generate more accurate price forecasts. In this article, a combination of deep learning algorithms is used to predict the prices of Bitcoin, Ethereum (ETH), and Monero (XMR). It is worth mentioning that based on the results of studies (Mansourabady et al., 2024; Tabe et al., 2024), it is better to use a combination of deep learning algorithms for this task because each deep learning algorithm alone, does not have the ability to discover important features well enough and can't solve the complexities present in the data.

Like other assets such as stocks and metals, the prices of cryptocurrencies are influenced by various factors, such as fake news, market manipulation attempts, and government policies (Zhang et al., 2021). Investing in highly volatile financial markets like cryptocurrencies can lead to significant profits and losses. These fluctuations can be rooted in news and emotions, making sentiment analysis important for predicting cryptocurrencies (Aslam et al., 2022). There is a vast amount of text in the world that can aid in the prediction of digital currencies. Although these texts contain a wealth of knowledge and information, it becomes increasingly difficult for humans to discover the necessary knowledge and information within the texts within specific time frames (Chowdhary, 2020). The goal of automated natural language processing is to efficiently and accurately perform this task faster and without human intervention. The rapid growth of internet-based applications, such as social media platforms and blogs, has led to an increase in the volume of news and opinions being generated (Wankhade et al., 2022). The demand for cryptocurrencies can be influenced by public opinion or government policies, resulting in fluctuations in their value (Aslam et al., 2022). In this regard, people's sentiments can contribute to determining the rise and fall of cryptocurrency market value. Sentiment analysis, especially for cryptocurrency investments, is currently highly popular. Studies have shown that news and tweets containing positive sentiments have a significant impact on the demand for cryptocurrencies, and the opposite holds true as well.

The main contribution of this study lies in the integration of Convolutional Neural Networks (CNN), Gated Recurrent Units (GRU), and dense layers, using historical data, along with the incorporation of a sentiment analysis component. To assess the impact of sentiment analysis on model performance, three well-established tools—VADER, TextBlob, and Flair—were applied individually, and their results were compared. The news data used for sentiment extraction were collected from Google News, providing a diverse and up-to-date source of market-related information. This comparative analysis offers a comprehensive understanding of the strengths and limitations of each sentiment analysis tool and contributes to enhancing the overall prediction accuracy. Model performance was evaluated using four standard metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Furthermore, comparisons with existing approaches in related studies demonstrate the superiority of the proposed method. Finally, applying the model to multiple cryptocurrencies, including BTC, ETH, and XMR, further confirms the robustness and reliability of the proposed algorithms. These findings highlight the effectiveness of combining deep learning and sentiment analysis for more reliable and data-driven cryptocurrency forecasting.

The next section reviews the previous works done in this field. The third section discusses the collected data and their pre-processing steps. The fourth section includes the theoretical foundations for understanding deep learning concepts and the proposed method. In the fifth section, the performance of the proposed model is analyzed in comparison with

other algorithms. Finally, in the conclusion section, findings and future work will be discussed.

2. Literature review

In this section, the previous works done to predict cryptocurrencies are reviewed. Some of the best previous studies models will be discussed in the next sections and are implemented for comparison.

(Lim et al., 2016) discussed that price prediction is used not only for cryptocurrencies but also for other market and fields. In this article, the prediction of housing prices in Singapore is discussed. Researchers collected timeseries data of various factors that influence housing prices, such as consumer consumption, wages, GDP, interest rates, population, and indicators related to housing and apartment prices. They used the data source of the Singapore Department of Statistics' real estate information system. In this study, ANN and ARIMA models were used to predict the future consumer price index (CPI). Different network architectures were experimented, and the best model was selected based on MSE performance. The results of this article indicate that the ANN model outperforms the ARIMA model in terms of prediction accuracy. This evaluation was performed using metrics such as MAE, RMSE, and MAPE, and according to the authors, their model has lower errors compared to the other models. In the next year, (Selvin et al., 2017) focused on predicting stock prices using deep neural networks. The article states that changes in the stock market may not always follow a regular pattern or have similar periods, so it is better to perform short-term predictions. The results also show that CNN has higher accuracy than Recurrent Neural Network (RNN), Long Short-Term Memory (LSTM), and ARIMA models for the collected data and has successfully predicted stock prices effectively.

(Struga & Irici, 2018) discussed the use of LSTM for Bitcoin price prediction. The authors briefly address Bitcoin and its impact on the world economy. They use a dataset consisting of market information, sentiments, blockchain data, and historical Bitcoin prices from the coinmarketcap website. They examine the prediction results for Bitcoin prices for the next 30 and 60 days. (Wimalagunaratne & Poravi, 2018) extending this line of research, suggested that cryptocurrency market is highly volatile, making it challenging to predict behavioral patterns. Therefore, in this article, authors focused on predicting the prices of cryptocurrencies using Twitter and historical data of cryptocurrencies. Three factors are considered for Bitcoin price prediction: sentiment analysis obtained through Twitter, transaction information, and historical cryptocurrency prices. For model testing, Bitcoin, Ethereum, and Bitcoin Cash are used. The results of a three-month period indicate that Ethereum predictions achieved higher accuracy with a score of 93 compared to the other two cryptocurrencies. In other works, (Pant et al., 2018) focused on predicting Bitcoin prices using historical and Twitter data, achieving an accuracy of 77.62. The method in this article involves first obtaining Twitter data and performing sentiment analysis on the tweets, accurately separating positive and negative tweets with an accuracy of 81.39. Then, this sentiment analysis is combined with historical data, and a prediction is made using a model consisting of one input layer, one GRU layer with 100 units, and one output layer. Furthermore, (Abraham et al., 2018) analyzes the impact of Bitcoin and Ethereum-related tweets. The analysis shows that tweets can have a direct or indirect influence on the market value of Bitcoin. The use of sentiment analysis of tweets and Google Trends data can serve as important inputs for modeling and price prediction. The results demonstrate a meaningful relationship between tweets and price changes. Additionally, the volume of tweets also fluctuates simultaneously with the prices of cryptocurrencies, increasing or decreasing accordingly.

(Rathan et al., 2019) used decision tree and linear regression algorithms to predict the price of Bitcoin for the next 5 days. According to the results, the linear regression algorithm achieved an accuracy of 97.5 %, which was higher than the decision tree algorithm with an accuracy of

Table 1

Bitcoin data taken from coinmarketcap website before preprocessing.

Open	High	Low	Close	Volume	Market Cap	Time Stamp
10328.73	10800.01	10680.84	3.55E+10	10266.01	1.97E+11	2020-09-14T23:59:59.999Z
10677.75	10938.63	10796.95	3.25E+10	10656.46	2.00E+11	2020-09-15T23:59:59.999Z
10797.76	11100.12	10974.9	3.08E+10	10704.88	2.03E+11	2020-09-16T23:59:59.999Z
10973.25	11037.42	10948.99	3.82E+10	10774.63	2.02E+11	2020-09-17T23:59:59.999Z
10951.82	11034.91	10944.59	2.63E+10	10829.66	2.02E+11	2020-09-18T23:59:59.999Z
10933.75	11134.09	11094.35	2.28E+10	10909.62	2.05E+11	2020-09-19T23:59:59.999Z
11095.87	11095.87	10938.27	2.47E+10	10814.48	2.02E+11	2020-09-20T23:59:59.999Z
10934.93	10988.3	10462.26	2.89E+10	10380.26	1.94E+11	2020-09-21T23:59:59.999Z
10459.62	10568.08	10538.46	2.36E+10	10382.73	1.95E+11	2020-09-22T23:59:59.999Z

95.8 %. Also, (Ji et al., 2019) investigated deep learning methods such as deep neural networks, long short-term memory models, convolutional neural networks, and residual neural networks and compared them for Bitcoin price prediction. The results of this study demonstrated that although long short-term memory models performed slightly better than other prediction models in regression mode for Bitcoin prices, deep neural network models exhibited the best performance in classification mode for predicting price increases and decreases.

(Poongodi et al., 2020) used a dataset covering the period from April 28, 2013, to July 31, 2017, and examined the Bitcoin price using an autoregressive integrated moving average model. The study achieved an accuracy of 49 %. Furthermore, the results indicated that removing trends and cycles improved the predictions significantly as timeseries models tend to capture regular patterns over time. (Patel et al., 2020) presented a hybrid approach for predicting the prices of cryptocurrencies based on LSTM and GRU is proposed. This article focuses solely on two cryptocurrencies, Litecoin and Monero, and uses daily data for prediction. The proposed algorithm of this article is based on the structure of recurrent neural networks, which can model sequential data. However, these networks face the issue of vanishing gradients, making their training challenging. To address this problem, both LSTM and GRU units are employed. The results demonstrate that this model outperforms the standalone LSTM algorithm and it is suggested to consider the influence of sentiment in future price predictions. (Derbentsev et al., 2020) discussed that to make investment decisions in the cryptocurrency market, it is better to have an assistant tool for price prediction. In this article, binary autoregressive tree algorithms, artificial neural networks, and ensemble decision trees were used to predict the prices of cryptocurrencies, including Bitcoin, Ethereum, and Ripple. According to the reported results, MAPE for binary autoregressive tree models and artificial neural networks is 3.5 %, and for the random forest model, it is less than 5 %. (Jiang, 2020) employed deep learning networks and various methods to improve the accuracy of Bitcoin price prediction. In this article, Bitcoin price data was collected and modified on an hourly basis to enable the prediction of Bitcoin prices at different time intervals. Different models were investigated, and the results indicated that the use of LSTM provided the best performance. Additionally, window-based normalization outperformed whole dataset normalization in terms of performance.

(Derbentsev et al., 2021) addresses the challenges of short-term prediction of cryptocurrencies using a supervised machine learning approach. For this purpose, two powerful ensemble methods, Random Forests (RF) and Stochastic Gradient Boosting Machines (SGBM) were employed. The dataset of this article consisted of Bitcoin, Ethereum, and Ripple currencies, and it used price and moving average as features. The results confirm that the ensemble machine learning approach is applicable for predicting cryptocurrency prices. The daily accuracy obtained by SGBM and RF, in terms of MAPE, for three currencies was between 0.92 % to 2.61 %. (Livieris et al., 2021) proposed a deep neural network model with multiple inputs for predicting the price and movement of cryptocurrencies, called MICDL. The data used in this research includes

daily historical data from January 1, 2017, to October 31, 2020, for Bitcoin, Ethereum, and Ripple. Experiments demonstrate that the proposed model in this article provides reliable predictions and outperforms traditional deep learning models.

(Aslam et al., 2022) focuses on sentiment analysis in cryptocurrencies and their price prediction. The authors of this article use relevant tweets for sentiment analysis and prediction, and preprocess the tweet data using techniques such as stemming, lemmatization, and removal of stop words. Their proposed model is a combination LSTM and GRU with features obtained from TF-IDF, word2vec, and bag-of-words. The results indicate that the proposed model has high accuracy in sentiment analysis and outperforms other machine learning and advanced models.

(Chen, 2023) aims to develop an algorithmic model with high prediction accuracy for the next-day price of Bitcoin using Random Forest Regression and LSTM. It also explores which variables impact the price of Bitcoin. In this study, descriptive variables such as Bitcoin price, specific technical features of Bitcoin, prices of other cryptocurrencies, commodities, market indices, exchange rates, and public attention were used for predicting the next-day price. Random Forest Regression outperformed LSTM in terms of price prediction accuracy. However, Random Forest Regression has drawbacks such as the inability to predict outcomes not observed in the training samples. Nevertheless, with an increase in Bitcoin transaction history, Random Forest Regression will have better performance in capturing Bitcoin price trends. (Murray et al., 2023) provides a comparative analysis of deep learning, machine learning, and statistical models for daily price prediction of cryptocurrencies. This study uses five popular cryptocurrencies, including Ripple, Bitcoin, Litecoin, Ethereum, and Monero. The results indicate that recurrent deep learning models perform the best. For example, LSTM achieved the best performance with a RMSE of 0.0222, and it also has lower training costs compared to other deep learning models. Additionally, simpler models such as GRU also demonstrated good performance. (Rathee et al., 2023) presents an improved approach using neural networks for long-term price prediction of cryptocurrencies. They proposed a CNN-BiLSTM model that uses both online and historical data for prediction. The proposed model consists of three convolutional layers and two bidirectional LSTM (Bi-LSTM) layers. MSE and RMSE are used for error evaluation. The proposed model achieved the lowest RMSE for Bitcoin online data and the historical dataset of Dogecoin. However, this approach does not consider important factors such as market sentiment.

(Samson, 2024) examines the use of three different machine learning algorithms, namely Random Forest, Gradient Boosting, and Bagging, for predicting the daily closing prices of six cryptocurrencies: Bitcoin, Binance, Ethereum, Solana, Dollar, and Ripple. The results showed that Random Forest Regression performed better than other machine learning algorithms for Binance, Ethereum, and Ripple, while Gradient Boosting demonstrated better performance for Bitcoin. This study shows that Random Forest Regression has superiority in predicting the closing prices of these cryptocurrencies.

Table 2

News data for “macroeconomic cryptocurrency” keyword.

Published Date	Title	Publisher Address
Mon, 04 Jan 2021 08:00:00 GMT	Mask Network Offers Tool for Twitter Token Sales – Crypto Briefing	https://cryptobriefing.com
Tue, 05 Jan 2021 08:00:00 GMT	Why Bitcoin Believers Could Be Proved Right In 2021 As The Bitcoin Price Continues To Climb – Forbes	https://www.forbes.com
Sat, 09 Jan 2021 08:00:00 GMT	Bitcoin Miners Saw 33 % Revenue Increase in December – CoinDesk	https://www.coindesk.com
Mon, 11 Jan 2021 08:00:00 GMT	This is how the bitcoin bubble will burst – Wired.co.uk	https://www.wired.co.uk
Mon, 11 Jan 2021 08:00:00 GMT	GemCoin Founder Sentenced to 10 Years for \$147 M Crypto Scheme – CoinDesk	https://www.coindesk.com
Thu, 14 Jan 2021 08:00:00 GMT	eToro warns users crypto trading may become limited due to ‘unprecedented demand’ – Cointelegraph	https://cointelegraph.com
Sat, 16 Jan 2021 08:00:00 GMT	Polkadot Price Analysis: DOT Skyrockets 30 % In 24 Hours, Is \$20 Next? – CryptoPotato	https://cryptopotato.com
Sun, 17 Jan 2021 08:00:00 GMT	65 % Say They Would Consider Selling Bitcoin If The Price Reaches \$100,000 – CryptoPotato	https://cryptopotato.com
Mon, 18 Jan 2021 08:00:00 GMT	Infinite Fleet publisher launches STO led by \$1M investment from Tether – Cointelegraph	https://cointelegraph.com

3. Data

The dataset used in this study comprises two components: historical cryptocurrency price data sourced from CoinMarketCap, and cryptocurrency-related news data collected from Google News.

3.1. Historical data

The first category of data comprises daily historical records for three cryptocurrencies—Bitcoin, Ethereum, and Monero—sourced from CoinMarketCap. Bitcoin (Bitcoin, 2008), introduced in 2008 by an individual or group under the pseudonym Satoshi Nakamoto, is the first and most widely recognized cryptocurrency. Ethereum (Wood, 2014), launched by Vitalik Buterin in 2014, is a decentralized blockchain platform and cryptocurrency. Monero, introduced in 2014, is a privacy-focused cryptocurrency emphasizing anonymity. The dataset for these cryptocurrencies includes daily features such as opening price, closing price, highest price, lowest price, trading volume, and market capitalization, covering the period from 2016 to September 2023. Table 1 provides a sample of the Bitcoin data.

3.2. News data

The sentiment analysis dataset was collected from Google News using Python web scraping libraries. Separate datasets were created for each cryptocurrency by extracting news articles using the corresponding cryptocurrency name as a keyword. The dataset spans from 2016 to September 2023, with a maximum of 24 news articles collected per day. Each entry includes the news title, publisher, publication date, and website URL. In addition, supplementary news articles were collected based on domain-relevant keywords, such as “bear market crypto,” “binance,” “bull market crypto,” “cryptocurrency GDP,” “inflation rates,” and “macroeconomic cryptocurrency,” with a limit of 50 articles per day. These articles were selected based on extensive literature review and field relevance. A sample of the news data is presented in Table 2.

3.3. Preprocess

Initially, redundant or less informative features were removed. Specifically, historical data features such as opening, highest, and lowest prices were excluded due to their high correlation with the closing price, which is commonly used in price prediction studies. For preprocessing, the remaining features were normalized to the range [0, 1] using the Min-Max scaling method implemented in the Scikit-learn library, as defined in formula 1. (Patro & Sahu, 2015).

$$x_{normalized} = \frac{x_{original} - x_{min}}{x_{max} - x_{min}} \quad (1)$$

Furthermore, to remove outliers, data points that account for more than 95 % or less than 5 % of the total data using Inter Quartile Range (IQR) have been replaced with the average of the preceding and succeeding data points to create a smooth and noise-free data point.

Based on prior research indicating that incorporating data from multiple previous days improves predictive accuracy (Jiang, 2020), a sliding window approach was applied. Specifically, to predict the next-day closing price, the closing prices from the previous 7 days were used as it yielded the highest prediction accuracy in the range of (1, 2, 3, 4, 5, 6, 7, 10, 14, 20, 30, 60, 100, 365).

3.4. Sentiment analysis

Since historical price data is numerical, it is necessary to convert textual news data into numerical form to enable effective feature extraction. One common approach to achieve this transformation is through sentiment analysis. In this study, three widely used sentiment analysis tools—VADER, TextBlob, and Flair—were applied individually to analyze news sentiments collected from Google. Employing these tools separately allows for a comprehensive and fair comparison of their performance. Moreover, this approach ensures that if one tool does not perform optimally on the dataset, its limitations do not significantly affect the overall assessment, thereby facilitating the selection of the most suitable sentiment analysis method for the proposed framework.

3.4.1. VADER

VADER (Valence Aware Dictionary and sEntiment Reasoner) combines lexical features with rule-based heuristics to generate a standardized sentiment lexicon, making it particularly suitable for short-text environments such as microblogs (Hutto & Gilbert, 2015). For sentiment analysis, the news dataset described previously is provided to VADER for each cryptocurrency, producing positive, negative, neutral, and compound sentiment scores. Daily sentiment features are then aggregated by averaging all news articles for a given day to align with the daily historical data. For days with no news, a value of −2 is assigned, outside the standard sentiment range of −1 to 1, based on experimental validation.

3.4.2. TextBlob

TextBlob facilitates various natural language processing tasks, including sentiment analysis and part-of-speech tagging (Bonta et al., 2019). Similar to VADER, the collected news articles were input into TextBlob to extract sentiment features. For each article, polarity (ranging from −1 for negative to 1 for positive sentiment, with 0 indicating neutral) and subjectivity (ranging from 0 for objective to 1 for subjective text) were computed.

3.4.3. Flair

Flair supports tasks such as named entity recognition, part-of-speech tagging, and multilingual sentiment analysis, including specialized support for medical text (Akbi et al., 2019). Prior studies have

demonstrated that Flair achieves competitive performance compared to other sentiment analysis tools and is widely adopted in industry. For each news article, Flair outputs two values: a sentiment label (positive or negative) and a sentiment intensity score ranging from 0 to 1 (Borrelli & Challiol, 2023).

3.5. Historical and sentiment data combination

After preprocessing and extracting sentiment scores, the historical and news datasets were merged. To preserve temporal order, a sequential index was included as a feature. The final input features for the model consist of the seven-day sliding window of closing prices, trading volume, market capitalization, sequential index, and aggregated sentiment measures.

4. Proposed method

The proposed method presented in this study consists of two main components. The first component focuses on predicting the next-day cryptocurrency price using a combination of deep learning architectures, while the second component integrates sentiment analysis derived from cryptocurrency-related news using tools like Flair, TextBlob, and Vader, as mentioned earlier. These two components are subsequently combined to form a unified and robust predictive framework for cryptocurrency price forecasting.

Algorithm1: Pseudocode

```

BEGIN
INPUT:
-Historical data of cryptocurrency: {date, close, open, high, low, volume,
marketcap}
-News data from Google News: {date, title, link}
PREPROCESSING:
1. Remove highly correlated features: open, high, low
2. Replace outliers with the average using the IQR
3. Normalize historical features using MinMaxScaler
4. Convert historical data into sliding windows of 7 and add a sequence number
5. Feed news data to VADER, TextBlob, and Flair for sentiment analysis
6. Calculate sentiment analysis for each day
7. Combine sentiment data with historical data
MODEL ARCHITECTURE:
1. 3 Conv1D Layers:
-Conv1D (128) -> Conv1D (64) -> Conv1D (128)
-Kernel Size: 5
2. 3 GRU Layers:
-GRU (256) -> GRU (256) -> GRU (256)
3. 1 Dense Layer:
-Dense (128, relu activation)
4. Output Layer:
-Dense (1, linear activation) → Predicted price
TRAINING:
-Loss function: Huber
-Optimizer: Adam
-Batch Size: 64
-Epochs: 1000
-Backpropagate errors and update weights
-Early stopping callback
-Reduce learning rate on plateau callback
EVALUATION:
-Metrics: MAE, MAPE, MSE, RMSE
-Validate on unseen time periods
PREDICTION:
-Predict the next day's closing price
END

```

As illustrated in Fig. 1, the workflow of the proposed model begins with the collection of historical cryptocurrency price data and corresponding news data from Google News. Sentiment analysis is then performed on the preprocessed news data using three widely recognized tools—VADER, TextBlob, and Flair—each applied individually. This step enables a comparative evaluation of the performance and impact of each sentiment analysis tool on prediction accuracy. The processed

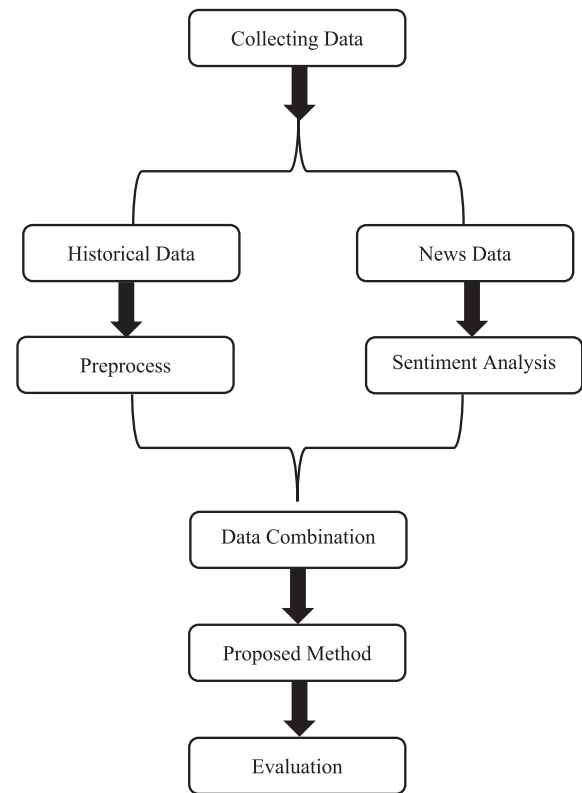


Fig. 1. Proposed Method.

sentiment data are then combined with historical price data and used as input for the proposed deep learning model. Finally, the trained model predicts the next-day price of cryptocurrencies. To enhance transparency and reproducibility, Algorithm 1 provides a detailed outline of the proposed approach.

4.1. Model

Two experimental scenarios are used for model evaluation. In the first scenario, only historical cryptocurrency data are used as input to the model. In the second scenario, both historical data and the sentiment information extracted from news are used. The sentiment analysis in the second scenario is conducted separately for VADER, TextBlob, and Flair, and the results are compared to determine which tool most effectively improves prediction performance.

The model is optimized using the Huber loss function, which is robust to outliers, and the Adam optimizer, known for its adaptive learning capability. During training, a batch size of 64 was used, and the model was run for a maximum of 1,000 epochs. To avoid overfitting and reduce unnecessary computation, an early stopping criterion with a patience value of 25 was incorporated into the callback function, allowing the training process to halt automatically when no further improvement was observed in the validation loss. The proposed deep learning architecture comprises three one-dimensional convolutional layers with 128, 64, and 128 filters, respectively, each employing a kernel size of 5. These convolutional layers are responsible for extracting spatial features from the input data. Following the convolutional layers, GRU layers are used, each with 256 units, to capture temporal dependencies within the time-series data. A fully connected dense layer with 128 neurons is then applied, followed by a final dense layer with a single neuron to output the predicted next-day price. The complete set of model hyperparameters is summarized in Table 3.

The detailed architecture of the proposed model is illustrated in Fig. 2. In the first scenario (historical data only), the input layer consists

Table 3
Hyperparameters for proposed method.

Hyperparameters	Values
Filters Conv1D	128, 64, 128
Kernel Size	5
Units GRU	256
Neurons Dense	128
Activation Function	Huber
Optimizer	Adam
Epochs	1000
Batch Size	64

of 10 features, including the closing prices from the previous seven days, the market value of the previous day, the trading volume of the previous day, and a sequential index. In the second scenario, when sentiment analysis is incorporated, additional input features are added depending on the sentiment analysis tool used:

- VADER: One additional feature column is added for each news keyword, representing the overall sentiment score, which yielded the most consistent results.
- Flair: Two additional columns are introduced for each news keyword, representing the sentiment class (positive/negative/neutral) and the corresponding sentiment intensity.
- TextBlob: Two additional columns for each news keyword are included to represent polarity and subjectivity scores.

In this architecture, which is shown in Fig. 2, the first layer serves as the input layer, followed by a reshaping operation to adapt the data for the Conv1D layers, which require three-dimensional input. The three convolutional layers (with 128, 64, and 128 filters) effectively extract spatial features. The subsequent GRU layers with 256 units each capture temporal patterns inherent in time-series data. The final dense layers synthesize these learned spatial and temporal representations, culminating in a single output neuron that predicts the next-day cryptocurrency price.

5. Results and comparison

In this section, the evaluation criteria for the implemented methods are discussed. Then, the implemented methods for a better comparison are reviewed. These methods include several deep learning algorithms and some models from previous articles to ensure a better comparison and thoroughly evaluate the performance of the proposed model. Finally, the obtained results for each cryptocurrency are examined. It should be noted that the necessary code for this study was written in Python version 3.11 and implemented using the PyCharm IDE. The system on which the codes were implemented had an Intel Core i7 12700 K processor, Nvidia GeForce RTX 3070 Ti graphics card, and 32 gigabytes of RAM.

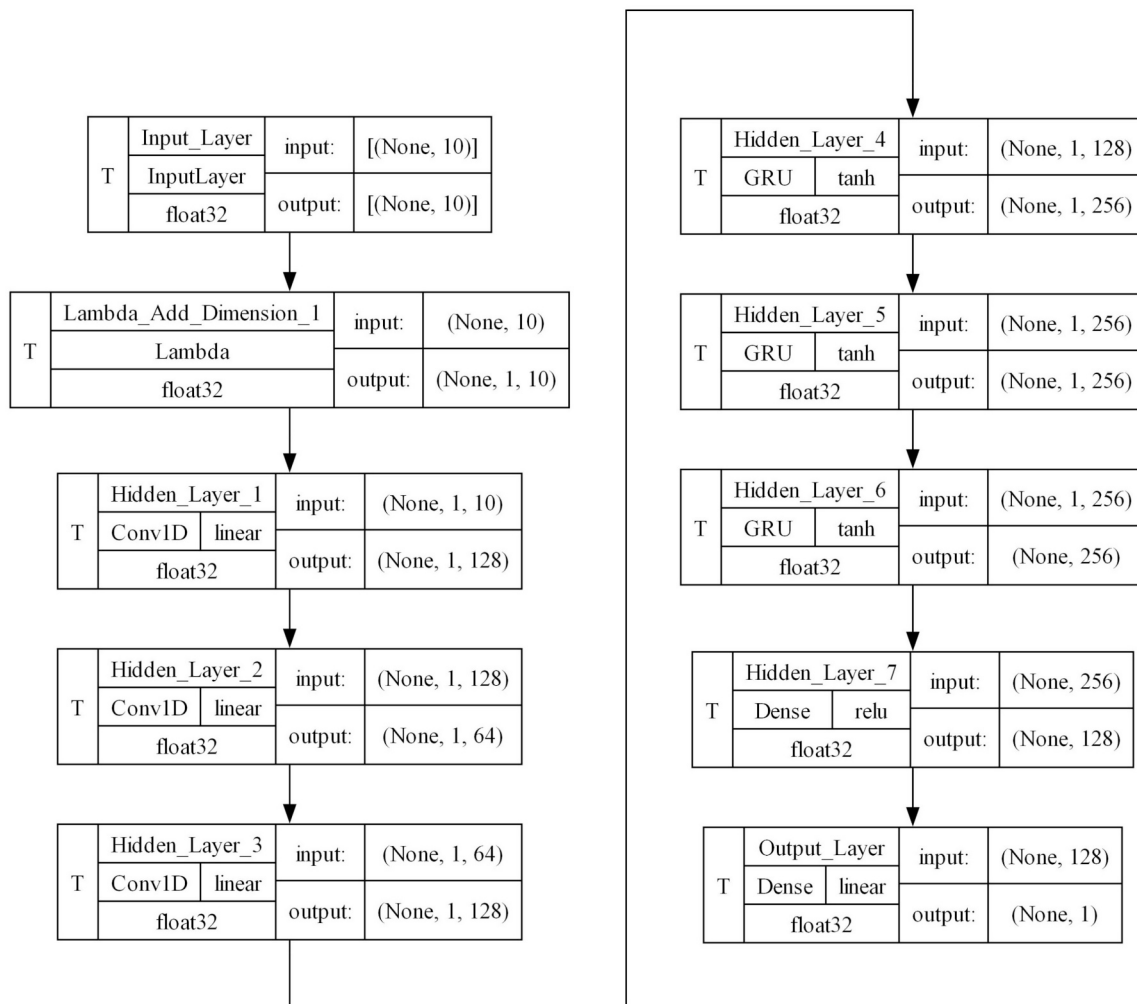


Fig. 2. Proposed Method.

5.1. Evaluation metrics

To evaluate deep learning models for predicting cryptocurrency prices, various metrics are commonly used. Below, some common metrics will be examined for evaluating regression models in the context of cryptocurrency price prediction:

MAE is the average of the absolute differences between the actual and predicted values, as indicated in equation (2) (Yao et al., 2018). MAE is used as a measure of accuracy for a model, and generally, lower values of this metric indicate better performance. This metric demonstrates how far the model's predictions are, on average, from reality. Unlike RMSE, the MAE is a more natural measure for the magnitude of the average error (Willmott & Matsuura, 2005).

$$MAE = \frac{\sum |y_i - \hat{y}_i|}{n} \quad (2)$$

MSE is one of the well-known types of error functions (Rathee et al., 2023). To evaluate using MSE, the difference between the actual value and the predicted value by the model is first calculated, then squared, and finally, the average of these squared differences is computed over all the data. This evaluation metric is shown in equation (3). The closer the value of the MSE is to zero, the better the model's performance in predicting the prices of cryptocurrencies. However, if the value of the MSE is large, it indicates that the model has errors in its price predictions. Therefore, the main objective in predicting the prices of cryptocurrencies is to minimize MSE. It is worth noting that MSE is very sensitive to outliers.

$$MSE = \frac{\sum (y_i - \hat{y}_i)^2}{n} \quad (3)$$

RMSE as shown in equation (4), indicates how well the regression line predicts the points (Rathee et al., 2023). It can also be interpreted as the standard deviation and, ideally, should be close to zero. In the presence of large values in the dataset, RMSE generally provides better results compared to the MSE.

$$RMSE = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n}} \quad (4)$$

MAPE is another common metric in regression tasks used to measure the accuracy of a predictive model, as shown in equation (5) (Zhang et al., 2021). This metric provides a percentage measure of the average absolute difference between the predicted values and the actual values. The absolute difference percentage for each data point is computed by taking the absolute difference between the predicted value and the actual value, then dividing it by the actual value, and finally multiplying by 100 to convert it to a percentage. The use of absolute value ensures that negative values are controlled in the summation. A lower value of MAPE indicates better model performance. However, it is important to note that MAPE has limitations, such as sensitivity to zero values in the actual data, as division by zero is undefined.

$$MAPE = \frac{1}{n} * \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| * 100 \quad (5)$$

In all four formulas above, y_i represents the actual value, $\hat{y}_i$ represents the predicted value by the model, and n represents the total number of data points.

5.2. Implemented models for comparison

To ensure that the proposed model performs well, several advanced deep learning algorithms have been implemented. These algorithms include Multi-Layer Perceptron (MLP) with 7 layers, CNN with 5 layers, RNN with 2 layers, Bidirectional RNN (Bi-RNN) with 2 layers, LSTM

with 2 layers Bi-LSTM with 2 layers, GRU with 2 layers, and Bidirectional GRU (Bi-GRU) with 2 layers. To ensure fairness, RNN, LSTM, and GRU models, which are all recurrent models, have the same hyperparameters. Similarly, Bi-RNN, Bi-LSTM, and Bi-GRU models, which are all bidirectional recurrent models, also have the same hyperparameters, as shown in Table 3. These hyperparameters have been selected based on achieving the best results on the data. For example, the MLP architecture performs best on the data when it has 7 consecutive layers, as shown in Table 4.

In order to compare the proposed model, in this article, seven methods from the reviewed papers have been implemented on the collected data to have a proper comparison. The first model is taken from (Rathee et al., 2023), which is a combination of three layers of CNN and two layers of Bi-LSTM. The second model is implemented based on the research in (Jiang, 2020), which is a combination of LSTM and a Feedforward Neural Network. The third method is implemented based on the study in (Murray et al., 2023), which is a combination of a 64-neuron CNN, a 75-neuron LSTM, and a 16-neuron Feedforward Neural Network. The fourth method is also from the same study, with a combination of a 75-neuron GRU and a 100-neuron Feedforward Neural Network. Additionally, the fifth model, consisting of two layers of LSTM, followed by a Feedforward Neural Network, and then a Gated Recurrent Unit, and finally a Feedforward Neural Network, is also from the same paper. For the next paper (Mazinani et al., 2025), the combination of one 128-neuron Dense, Multi-Head Attention with 4 attention heads and 128-dimensional keys, and 2 dense layers consisting of 128 neurons is used. As for the last model (Amirshahi & Lahmirmi, 2025), a more advanced architecture is employed, consisting of one 64-neuron CNN, two LSTMs with 50 and 100 units respectively, an attention layer with 64 units, and one 128-neuron dense layer. These models and their references can be seen in Table 5.

The preprocessed data is divided into two categories: train set and test set. Specifically, 90 % of the data is used for training, and the last 10 % is reserved for testing. Since the data is timeseries, basic methods from the Scikit-learn library in Python cannot be used for data splitting, as the

Table 4
Deep learning models for comparison.

Algorithms	Layers (Neuron – Filter – Kernel – Activation Function)
MLP	Dense (256 – ReLU), Dense (256 – ReLU), Dense (512 – ReLU), Dense (1024 – ReLU), Dense (512 – ReLU), Dense (256 – ReLU), Dense (256 – ReLU)
CNN	Conv1D (32–3 – ReLU), Conv1D (64–3 – ReLU), Conv1D (128–3 – ReLU), Conv1D (64–3 – ReLU), Conv1D (32–3 – ReLU)
RNN	SimpleRNN (128 – ReLU), SimpleRNN (128 – ReLU)
Bi-RNN	BiDirectional(SimpleRNN (64 – ReLU)) with recurrent-dropout(0.1), BiDirectional(SimpleRNN (64 – ReLU)) with recurrent-dropout(0.05)
LSTM	LSTM (128 – ReLU), LSTM (128 – ReLU)
Bi-LSTM	BiDirectional(LSTM (64 – ReLU)) with recurrent-dropout(0.1), BiDirectional(LSTM (64 – ReLU)) with recurrent-dropout(0.05)
GRU	GRU (128 – ReLU), GRU (128 – ReLU)
Bi-GRU	BiDirectional(GRU (64 – ReLU)) with recurrent-dropout(0.1), BiDirectional(GRU (64 – ReLU)) with recurrent-dropout(0.05)

Table 5
Implemented models of previous studies with resources.

Source	Proposed Model
(Rathee et al., 2023)	3Conv1D (64, 128, 64) + 2Bi-LSTM (100)
(Jiang, 2020)	2LSTM (256) + Dense (128)
(Murray et al., 2023)	Conv1D (64) + LSTM (75) + Dense (16)
(Murray et al., 2023)	GRU (75) + Dense (100)
(Murray et al., 2023)	LSTM (75) + LSTM (50) + Dense (32) + GRU (50) + Dense (64)
(Mazinani et al., 2025)	Dense (128) + MultiHeadAttention (4, 128) + 2Dense (128)
(Amirshahi & Lahmirmi, 2025)	Conv1D (64) + 2LSTM (50, 100) + Attention (64) + Dense (128)

order of the data is important. Therefore, the Python slicing method is employed to preserve the order, and the last 10 % is used as the testing data.

5.3. BTC results

For evaluating the implemented models, historical Bitcoin data has been provided as input to the first 16 models of Table 6. This data includes features such as the closing price of the previous 7 days, the trading volume of the previous day, the market capitalization of the previous day, and a sequence of numbers for predicting the closing price of the current day. Unlike most previous research where only the closing price of the previous day was given, in this paper, the price of the past 7 days is provided as input. This approach has resulted in higher model accuracy. Therefore, the closing price data is practically incorporated into the proposed approach as a window. In Table 6, models 17 to 19, the additional features that have been incorporated into the proposed approach are the sentiment analysis data from Vader, Flair, and TextBlob tools which include news articles with the tags “bitcoin,” “bear market crypto,” and “bull market crypto”. The best results have been obtained by combining the sentiment analysis results of these news articles. As can be seen in Table 6, rows 1 to 8 represent advanced deep learning algorithms, rows 9 to 15 represent proposed models from previous studies, and rows 16 to 19 represent the proposed approach with different data for the Bitcoin cryptocurrency. In this table, four evaluation metrics have been included to better examine and compare the models. Among the models 1 to 8, which represent advanced deep learning algorithms, GRU performs the best in both the MAE and MSE metrics, demonstrating superior performance. Using only historical Bitcoin data, the proposed algorithm from Study (Rathee et al., 2023) which is in row 9 shows the best performance in terms of MAE, while the proposed algorithm from Paper (Mazinani et al., 2025) which is in row 14 demonstrates the best performance in terms of MSE. It is worth noting that the MAE and MSE of the proposed approach, when only historical data is used as input, outperform all implemented models. When sentiment analysis of news is added, the accuracy improves even more. According to the results in Table 6, the best accuracy is achieved by the

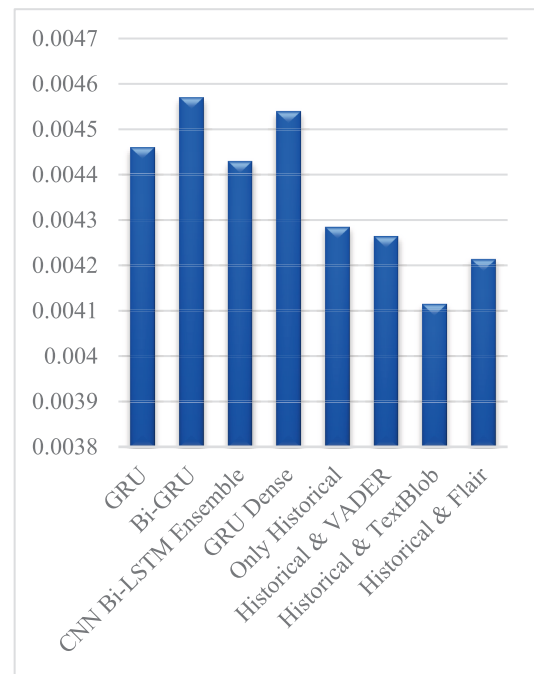


Fig. 3. MAE comparison for Bitcoin.

proposed approach using sentiment analysis data generated from TextBlob. The proposed approach with Flair ranks second, followed by the proposed approach with VADER.

The figures above compare the MAE, MSE, MAPE, and RMSE for the top-performing models of each category. In Fig. 3, which compares the top two models from each category based on the MAE metric, it can be concluded that as the model becomes more complex, the accuracy increases. Similarly, in Figs. 4, 5, and 6, which compare the MSE, MAPE, and RMSE, respectively, the same result can be observed that the accuracy improves with the increasing complexity of the architecture.

Table 6

Comparison of implemented models for Bitcoin cryptocurrency.

Category	Number	Algorithms	MAE	MAPE	MSE	RMSE
Advanced Deep Learning Models	1	MLP	0.00484	1.1731	0.00005735	0.00757
	2	CNN	0.00535	1.3046	0.00006127	0.00782
	3	RNN	0.00460	1.1185	0.00005594	0.00747
	4	Bi-RNN	0.00477	1.1609	0.000054441	0.00737
	5	LSTM	0.00486	1.1844	0.00005544	0.00744
	6	Bi-LSTM	0.00459	1.1176	0.00005429	0.00736
	7	GRU	0.00446	1.0938	0.00005287	0.00727
	8	Bi-GRU	0.00457	1.1129	0.00005399	0.00734
Reviewed Papers Models	9	CNN Bi-LSTM Ensemble (Rathee et al., 2023)	0.00443	1.1243	0.00005700	0.00755
	10	2LSTM DENSE (Jiang, 2020)	0.00497	1.2101	6960.00005	540.007
	11	CNN LSTM DENSE (Murray et al., 2023)	0.00478	1.2183	0.00005654	0.00751
	12	GRU DENSE (Murray et al., 2023)	4540.00	1.1049	54930.0000	0.00741
	13	2LSTM DENSE GRU Dense (Murray et al., 2023)	0.00462	1.1753	0.00005583	0.00747
	14	Dense (128) + MultiHeadAttention (4, 128) + 2Dense (128) (Mazinani et al., 2025)	0.00462	1.1227	0.00005448	0.00738
	15	Conv1D (64) + 2LSTM (50, 100) + Attention (64) + Dense (128) (Amirshahi & Lahmiri, 2025)	0.00512	1.2438	0.00005846	0.00764
Proposed Models	16	Proposed Method (Only Historical)	0.004285	1.0888	0.000051445	0.00717
	17	Proposed Method (Historical and Sentiment with VADER)	0.004264	1.0837	0.000052101	0.00721
	18	Proposed Method (Historical and Sentiment with TextBlob)	0.004115	1.0465	0.000051797	0.00719
	19	Proposed Method (Historical and Sentiment with Flair)	0.004214	1.0753	0.000049262	0.00701

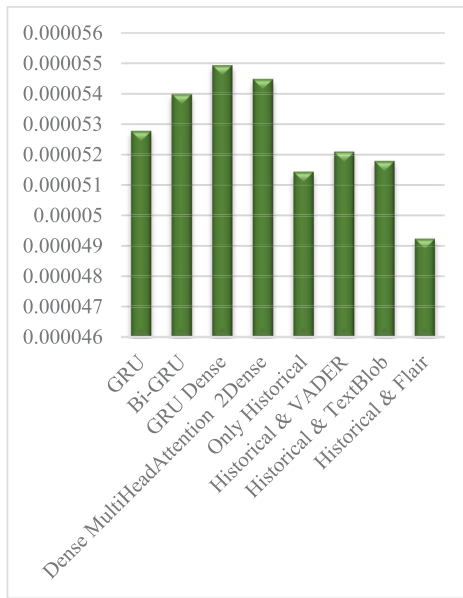


Fig. 4. MSE comparison for Bitcoin.

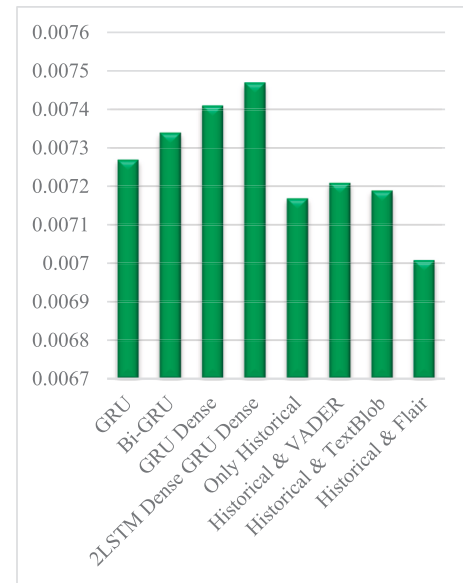


Fig. 6. RMSE comparison for Bitcoin.

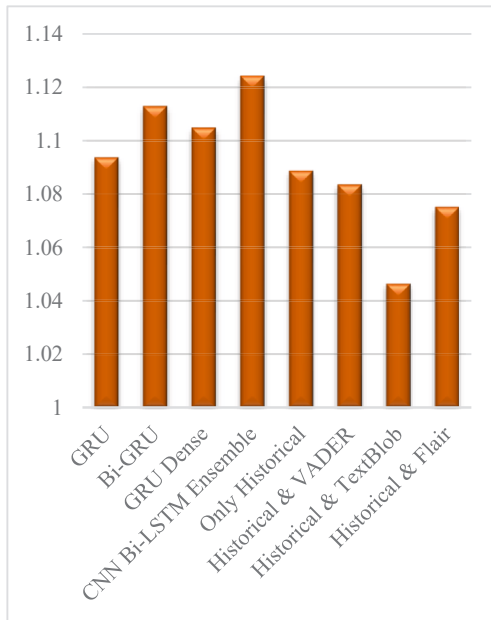


Fig. 5. MAPE comparison for Bitcoin.

According to MAPE and RMSE metrics, GRU achieves the best results. Ultimately, as can be seen in Table 6, the best results are obtained from the proposed approach of the paper. It is worth noting that in both cases, when only historical data is used as input and when sentiment analysis data from news is also added, the best approach is the proposed model from this paper. Contrary to MAE and MAPE, where the best results are achieved with TextBlob sentiment analysis, in terms of MSE and RMSE, the best sentiment analysis tool is Flair. This is because MSE squares the error values and gives more weight to larger errors.

5.4. ETH results

The feature vector provided as input to the algorithms for predicting the next day's price of Ethereum in the historical price data includes the closing price of the past 7 days, the trading volume of the previous day, the market cap of the previous day, and a sequence of numbers just like Bitcoin. Sentiment analysis was conducted on news articles tagged with "bitcoin", "ethereum", "bear market crypto," and "bull market crypto." The results of implemented models on Ethereum can be seen in Table 7. According to this table, among advanced deep learning models, LSTM achieved the best results in terms of MAE, while GRU performed better when considering MSE. The best-performing approach, based on the MAE metric, was found to be model 14, corresponding to the article (Jiang, 2020). This method is essentially a combination of two LSTM layers and one dense layer. Similar to the results obtained for Bitcoin, in this case as well, when only historical Ethereum data is provided to the proposed approach, it outperforms all other models. Furthermore, when news data is added, the accuracy improves, and the best results are obtained from sentiment analysis using TextBlob. After TextBlob, the combined approach of Flair and then VADER achieves the best results.

As seen in Figs. 7, 8, 9, and 10, four metrics, including MAE, MSE, MAPE, and RMSE, are compared for Ethereum cryptocurrency. Regarding advanced deep learning algorithms, for the MAE error metric, RNN had a lower error compared to the other models. Finally, the best-performing approach among the reviewed papers based on the MSE metric was the study (Jiang, 2020), although its error is slightly higher than the proposed approach in this article. The same results also hold for RMSE. According to the figures above, the superiority of the proposed models in this article can be observed in all metrics.

5.5. XMR results

Similar to the previous sections, the feature vector consists of the Monero closing price in 7-days format, the trading volume of the previous day, the market capitalization of the previous day, and a sequence of numbers. The news tags used are "cryptocurrency gdp" and

Table 7
Comparison of implemented models for Ethereum cryptocurrency.

Category	Number	Algorithms	MAE	MAPE	RMSE	MSE
Advanced Deep Learning Models	1	MLP	0.00644	1.7764	0.00838	0.000070283
	2	CNN	0.00600	1.6437	0.00790	0.000062416
	3	RNN	0.00559	1.5134	0.00777	0.000060522
	4	Bi-RNN	0.00672	1.7764	0.00899	0.000080992
	5	LSTM	0.00566	1.5527	0.00772	0.000059675
	6	Bi-LSTM	0.00566	1.5515	0.00775	0.000060108
	7	GRU	0.00867	1.3070	0.01053	0.000110957
	8	Bi-GRU	0.00858	1.6015	0.00785	0.000061734
Reviewed Papers Models	9	CNN Bi-LSTM Ensemble (Rathee et al., 2023)	0.00749	2.1619	0.00867	0.000075233
	10	2LSTM DENSE (Jiang, 2020)	0.00495	1.4272	0.00665	0.000044233
	11	CNN LSTM DENSE (Murray et al., 2023)	0.00526	1.4386	0.00746	557820.0000
	12	GRU DENSE (Murray et al., 2023)	0.00617	1.7756	0.00741	0.000055000
	13	2LSTM DENSE GRU Dense (Murray et al., 2023)	0.00588	1.6110	0.00786	0.000061851
	14	Dense (128) + MultiHeadAttention (4, 128) + 2Dense (128) (Mazinani et al., 2025)	0.00659	1.8910	0.00774	0.00005997
	15	Conv1D (64) + 2LSTM (50, 100) + Attention (64) + Dense (128) (Amirshahi & Lahmirm, 2025)	0.006992	2.0215	0.00838	0.00007022
Proposed Models	16	Proposed Method (Only Historical)	0.00486	1.4106	0.00664	0.000044213
	17	Proposed Method (Historical and Sentiment with VADER)	0.00473	1.3683	0.00675	0.000045646
	18	Proposed Method (Historical and Sentiment with TextBlob)	0.00419	1.2169	0.00641	0.000041108
	19	Proposed Method (Historical and Sentiment with Flair)	0.00441	1.2723	0.00632	0.000040041

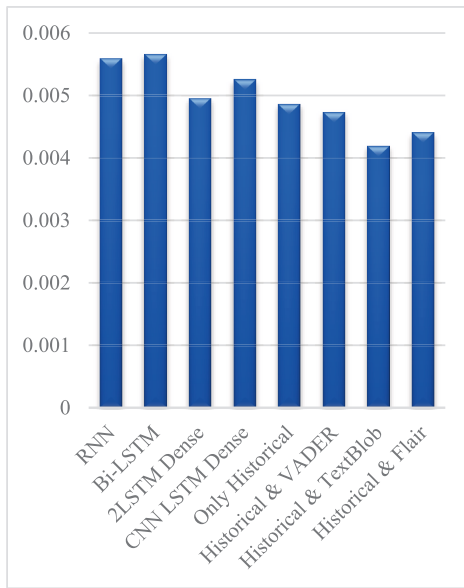


Fig. 7. MAE comparison for Ethereum.

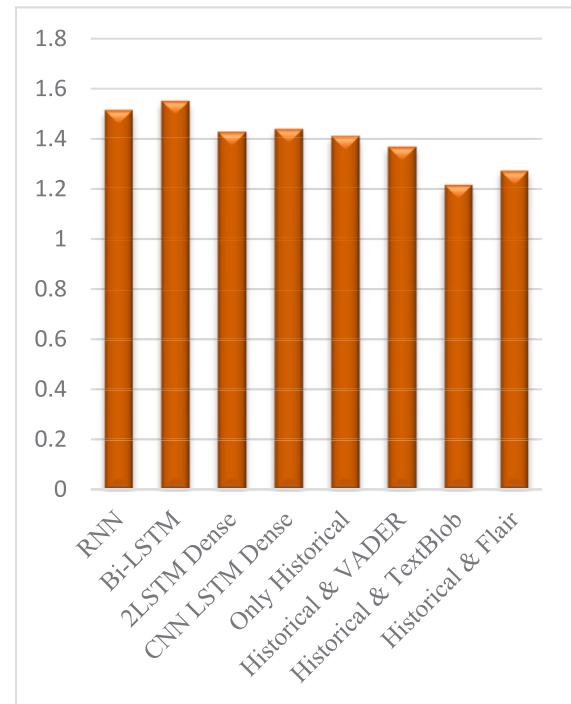


Fig. 9. MAPE comparison for Ethereum.

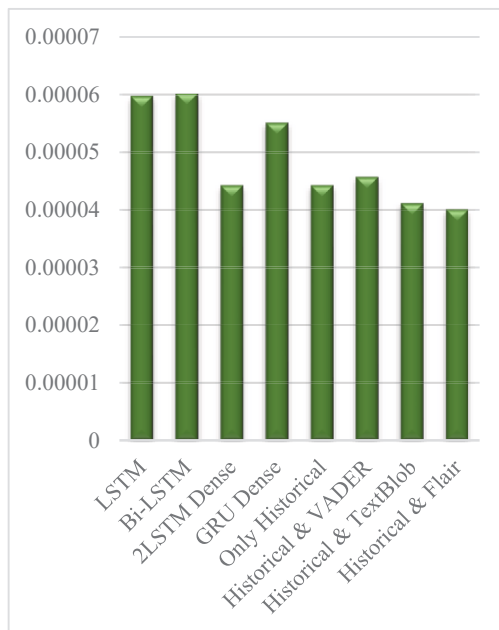


Fig. 8. MSE comparison for Ethereum.

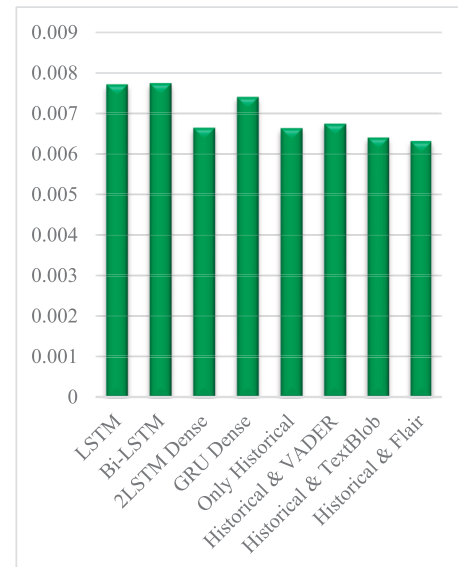


Fig. 10. RMSE comparison for Ethereum.

“macroeconomics cryptocurrency”. Due to its different nature compared to Ethereum and Bitcoin, Monero is less sensitive to news. This cryptocurrency primarily focuses on transactional security and anonymity, which attracts a different set of investors and the target community. According to conducted experiments, only news related to the mentioned tags had a positive impact on the results.

The results of Monero cryptocurrency are presented in Table 8. Among advanced deep learning algorithms, the top-performing model was LSTM. When previous papers' models were tested on the data, the

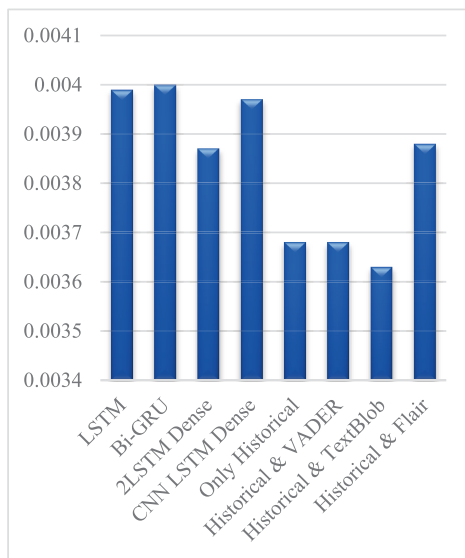
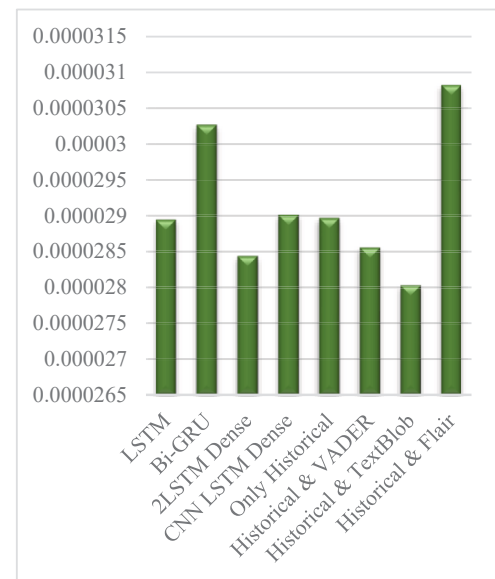
best result was obtained from the article (Jiang, 2020), which is listed in row 10. Similar to the results obtained from Bitcoin and Ethereum, MAE is better when only historical Monero data is used as input for the proposed approach compared to all the other implemented algorithms. Additionally, when sentiment analysis data is added as input, the results improve similarly to other cryptocurrencies. In this case, the best result is obtained from the TextBlob tool, followed by VADER, and finally Flair.

LSTM and GRU have also shown good results the advanced deep learning category. In comparison to previous studies and according to

Table 8

Comparison of implemented models for Monero cryptocurrency.

Category	Number	Algorithms	MAE	MAPE	MSE	RMSE
Advanced Deep Learning Models	1	MLP	0.00432	1.39061	0.000034177	0.00584
	2	CNN	0.00475	1.51250	0.000038372	0.00619
	3	RNN	0.00414	1.34747	0.000033053	0.00574
	4	Bi-RNN	0.00410	1.31337	0.000033393	0.00577
	5	LSTM	0.00399	1.28319	0.000028954	0.00538
	6	Bi-LSTM	0.00413	1.33001	0.000033127	0.00575
	7	GRU	0.00435	1.38366	0.000033799	0.00581
	8	Bi-GRU	0.00400	1.28740	0.000030270	0.00550
Reviewed Papers Models	9	CNN Bi-LSTM Ensemble (Rathee et al., 2023)	0.00450	1.43559	0.000034459	0.00587
	10	2LSTM DENSE (Jiang, 2020)	0.00387	1.24825	0.000028447	0.00533
	11	CNN LSTM DENSE (Murray et al., 2023)	0.00397	1.28751	0.000029019	0.00538
	12	GRU DENSE (Murray et al., 2023)	0.00406	1.30652	0.000029402	0.00542
	13	2LSTM DENSE GRU Dense (Murray et al., 2023)	0.00433	1.39575	0.000032201	0.00567
	14	Dense (128) + MultiHeadAttention (4, 128) + 2Dense (128) (Mazinani et al., 2025)	0.00393	1.32219	0.000033812	0.00581
	15	Conv1D (64) + 2LSTM (50, 100) + Attention (64) + Dense (128) (Amirshahi & Lahmiri, 2025)	0.00389	1.25705	0.000028776	0.00536
Proposed Models	16	Proposed Method (Only Historical)	0.00368	1.23367	0.000028975	0.00538
	17	Proposed Method (Historical and Sentiment with VADER)	0.00368	1.23401	0.000028565	0.00534
	18	Proposed Method (Historical and Sentiment with TextBlob)	0.00363	1.21204	0.000028037	0.00529
	19	Proposed Method (Historical and Sentiment with Flair)	0.00388	1.30247	0.000030825	0.00555

**Fig. 11.** MAE comparison for Monero.**Fig. 12.** MSE comparison for Monero.

the MAPE metric, the article (Jiang, 2020) with 2 LSTMs and the article (Murray et al., 2023) with a combination of CNN, LSTM, and a dense layer have achieved the best results. Based on the RMSE metric results for Monero shown in Figs. 11–14, the same outcome as Bitcoin and Ethereum is achieved. Similar to the previous two cryptocurrencies, the best result was obtained from the proposed approach with a combined input of historical data and news sentiment using TextBlob. It is also worth mentioning that, based on the following figures, if only historical price data is provided to the proposed approach, it will still have lower errors compared to all the other implemented models.

In this section, the results of the implemented algorithms and the proposed approach for analyzing cryptocurrencies such as Bitcoin, Ethereum, and Monero have been examined. According to the tables and figures, the superiority of the proposed approach was evident, and the

impact of sentiment analysis was also seen positively. Overall, TextBlob was recognized as the best tool for sentiment analysis in this study. As observed, GRU had lower error rates in most cases, which is why it was used in the proposed model. Additionally, two of the top proposed architectures from previous studies were as follows: Article (Hull, 2009), which had 2 LSTMs with a dense layer, and Article (Rathee et al., 2023), which included a GRU. Overall, the results show that combining CNN and GRU can help the model to predict more effectively because the model can extract both spatial and temporal features. The next section will cover the general conclusions and provide recommendations for future research. Of course, it should be noted that the articles (Amirshahi & Lahmiri, 2025; Mazinani et al., 2025) that use attention models have also shown very good results and may have better results if they are given news data in addition to historical data.

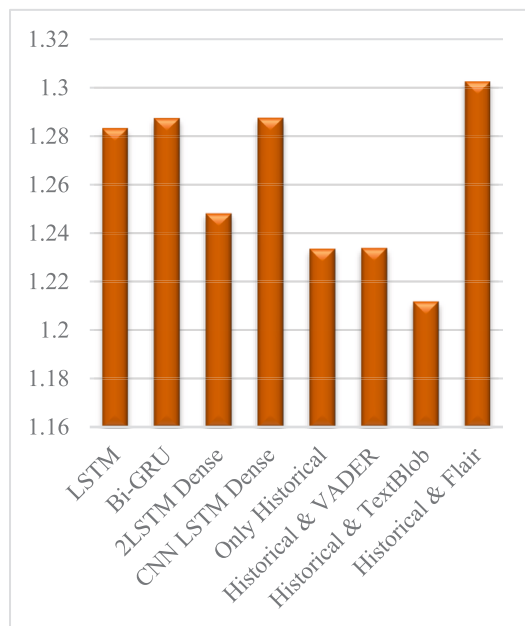


Fig. 13. MAPE comparison for Monero.

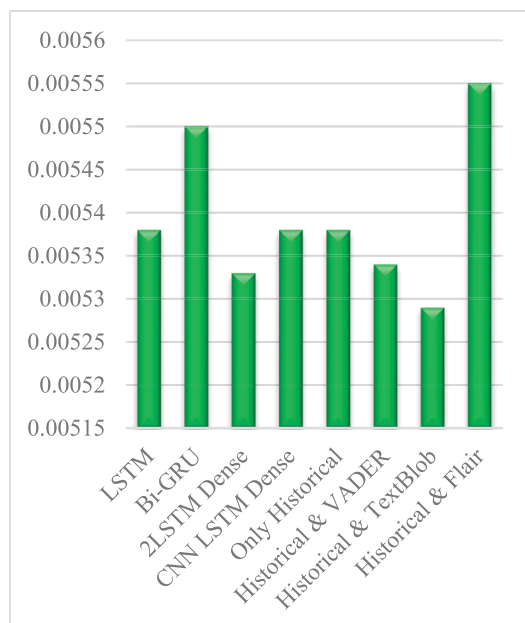


Fig. 14. RMSE comparison for Monero.

6. Conclusion and future work

This study presented a hybrid deep learning model combining CNN and GRU with sentiment analysis to predict cryptocurrency prices. The results showed that integrating historical data with news sentiment significantly improved prediction accuracy for Bitcoin, Ethereum, and Monero. Using a 7-day window of past prices was particularly effective in capturing market patterns, while each cryptocurrency displayed its own unique behavior. Although the proposed method outperformed existing approaches, further improvements can be achieved by

expanding datasets, incorporating additional financial indicators, and reducing prediction intervals to shorter time frames. Future work should also consider alternative sentiment sources, such as Twitter, and employ more advanced text analysis techniques. Overall, the findings confirm that combining deep learning with sentiment analysis provides a promising direction for enhancing cryptocurrency price prediction.

CRedit authorship contribution statement

Fatemeh Tabe: Conceptualization, Methodology, Software, Formal analysis, Validation, Formal analysis, Resources, Investigation, Visualization. **Ali Mansourabady:** Conceptualization, Methodology, Software, Formal analysis, Validation, Formal analysis, Resources, Investigation, Visualization. **Amir Hossein Rasekh:** Conceptualization, Validation, Formal analysis, Investigation, Supervision. **Betsabeh Tanoori:** Conceptualization, Validation, Formal analysis, Investigation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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